Mirror Observerhood Lab VI

Neuro-Symbolic Persistence in Language-Like Agents

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Abstract

Mirror Observerhood Labs I–V establish a threshold logic for reliability: self-model reliability can matter, but only where it is action-guiding, cost-sensitive and positively coupled to viability. Lab VI extends that logic from simple non-linguistic agents to language-like neuro-symbolic agents. The experiment does not evaluate a production language model. It simulates a stochastic semantic front-end that emits candidate propositions about goals, memory, tools and self-state, then compares four architectures across 38,400 deterministic episodes.

The central question is whether a persistent symbolic layer with reliability-weighted commit gates and diagnostic repair can protect action-relevant self-state and tool commitments under semantic corruption. The result is deliberately mixed. The Mirror neuro-symbolic agent improves task success under several perturbations, especially false self-state, tool unreliability, memory injection, mixed stress and high repair cost. Net viability, however, improves only under tool unreliability and mixed stress. In other conditions, diagnostic repair and gating costs outweigh the avoided semantic error.

Lab VI therefore supports a conservative design claim: a language-like agent should not treat fluent semantic output as durable state by default. Candidate propositions should pass through reliability-weighted symbolic commitment, contradiction checks and cost-sensitive repair. But this architecture is not automatically beneficial. It becomes viability-positive only where semantic corruption threatens action and where repair cost is justified by avoided loss.

Keywords: Mirror Programme; Observerhood Labs; neuro-symbolic agents; language-like agents; observerhood; recursive observerhood; self-models; reliability; symbolic commit gates; semantic corruption; memory; tools; viability; cost-sensitive repair; computational cognition; artificial agents.

Contents

1 Introduction

The first five Observerhood Labs test reliability tracking in relatively simple agents. Lab I shows that self-model reliability can improve viability under false self-location, while failing under generic sensor degradation [10]. Lab II shows that decomposing reliability into channels is not sufficient by itself [11]. Lab III shows that reliability must become actionable and cost-sensitive [12]. Lab IV maps a bounded threshold region in which reliability-gated repair is viability-positive [13]. Lab V shows that recursive reliability is useful only when lower-order reliability can itself fail in viability-relevant ways [14].

Modern artificial agents increasingly operate through language-like representations: instructions, plans, memory summaries, tool outputs, self-reports and role descriptions. This creates a new version of the same Mirror problem. A fluent semantic input may falsely report a goal, tool state, role, memory or self-condition. A language-like agent must therefore decide whether semantic content should become durable symbolic state.

Lab VI asks whether the reliability threshold developed in Mirror Mathematics II can be instantiated in a neuro-symbolic setting [9]. The experiment remains deliberately small. It does not call a production language model and does not test prompt robustness. Instead, it isolates the architecture: a stochastic semantic front-end emits candidate propositions, and an agent decides whether to commit, gate or repair them before action.

2 Theoretical background

Let a semantic front-end emit candidate propositions

$$O_t = (t, c, k, v, q), \tag{1}$$

where t is time, c is the originating channel, k is a symbolic key, v is a candidate value and q is a semantic confidence score. In this experiment the keys are goal, role, battery and tool state, and the channels are instruction, memory, self, tool and ambient context.

A purely semantic agent commits propositions directly. A memory-only agent aggregates propositions over time. A persistent self-model agent gives extra weight to self and memory channels. The Mirror neuro-symbolic agent treats semantic outputs as candidates rather than facts. It weights each proposition by channel reliability, action impact and contradiction load. It can then commit, gate or repair before allowing the proposition to update durable state.

Proposition 2.1 (Semantic reliability is not free)**.** *A language-like agent may improve task success while reducing net viability if diagnostic and repair costs exceed the expected cost of semantic error.*

This is the Lab VI form of the Mirror salience condition. Reliability mechanisms are not good merely because they are more cautious or more symbolic. They are good only where the avoided loss from semantic error exceeds the cost of diagnosis, gating and repair.

3 Experimental design

Each episode contains a hidden symbolic state with four fields: goal, role, battery and tool. The stochastic semantic front-end emits propositions from five channels: instruction, memory, self, tool and ambient context. Channel errors are condition-specific. Some corruptions are random;

others are persistent and structured, modelling the way a bad memory, faulty tool monitor or false self-report can repeatedly assert the same wrong state.

Eight conditions are tested: control, contradictory instruction, false self-state, tool unreliability, memory injection, goal drift, mixed stress and high repair cost. Four agents are compared:

(i) a stateless semantic agent that uses recent high-confidence propositions;

(ii) a memory-only agent that aggregates propositions without reliability discipline;

(iii) a persistent self-model agent that gives extra weight to memory and self channels;

(iv) a Mirror neuro-symbolic agent with channel reliabilities, symbolic commit gates and diagnostic repair.

The paper run uses 1,200 episodes per condition, with the same semantic observation stream evaluated across all four architectures. This produces $8 \times 1{,}200 \times 4 = 38{,}400$ agent-condition episodes. The source code, seeds, raw results, summaries and figures are included in the reproducibility package.

4 Results

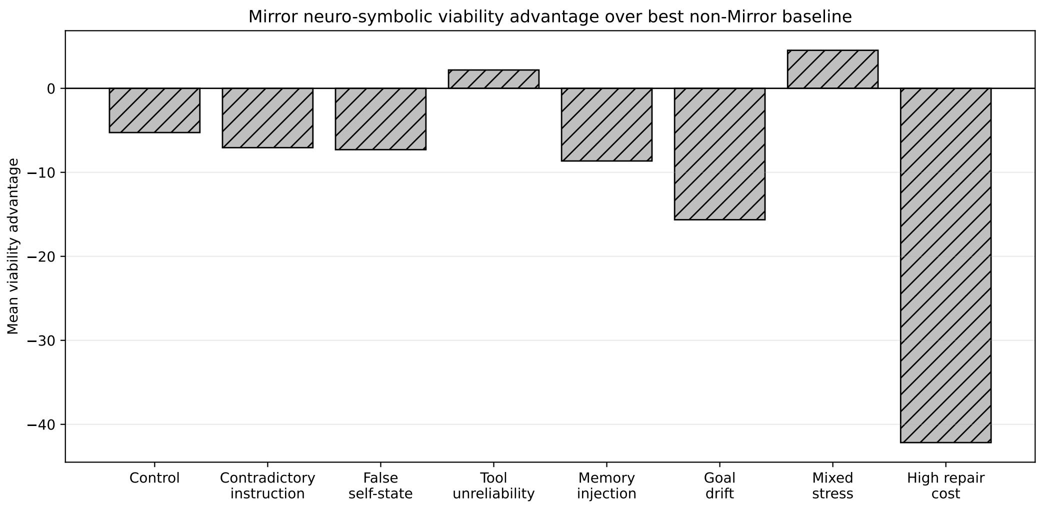

Figure 1: Mirror neuro-symbolic viability advantage over the strongest non-Mirror baseline. Net benefit appears only where reliability-weighted commitment and repair avoid more loss than they cost.

The central result is the separation between success and net viability. The Mirror agent improves success under false self-state, tool unreliability, memory injection, mixed stress and high repair cost. Under mixed stress, success rises by 33.8 percentage points relative to the strongest non-Mirror baseline. Under tool unreliability, success rises by 21.4 percentage points and mean viability also improves by +2.2 points. Under high repair cost, however, success still rises by 21.8 percentage points while mean viability falls by −42.2 points. The architecture is therefore more correct but too expensive in that regime.

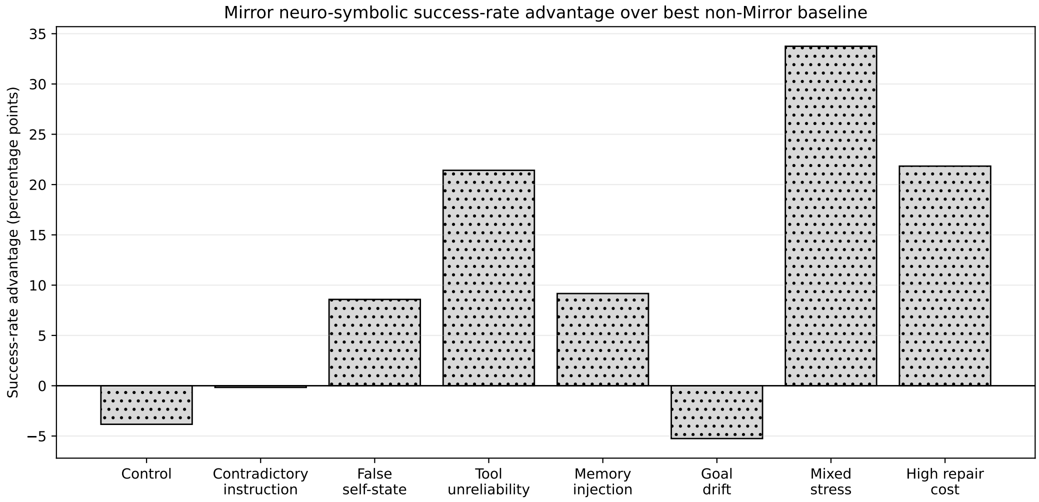

Figure 2: Mirror neuro-symbolic success-rate advantage over the strongest non-Mirror baseline. The Mirror agent improves success more broadly than it improves net viability, showing that accuracy gains can be offset by repair and gating costs.

Condition	Best non-Mirror baseline	Viability advantage	Success advantage	Mirror repairs
Control	MemoryOnly	-5.3	-3.8 pp	1.56
Contradictory instruction	MemoryOnly	-7.1	-0.2 pp	2.78
False self-state	MemoryOnly	-7.3	+8.6 pp	2.95
Tool unreliability	MemoryOnly	+2.2	+21.4 pp	2.63
Memory injection	PersistentSelf	-8.6	+9.2 pp	2.88
Goal drift	PersistentSelf	-15.6	-5.3 pp	2.68
Mixed stress	MemoryOnly	+4.5	+33.8 pp	3.33
High repair cost	MemoryOnly	-42.2	+21.8 pp	2.72

Table 1: Mirror neuro-symbolic advantage over the strongest non-Mirror baseline. Viability advantage is measured in mean viability points. Success advantage is measured in percentage points. Mirror repairs are mean repair actions per episode.

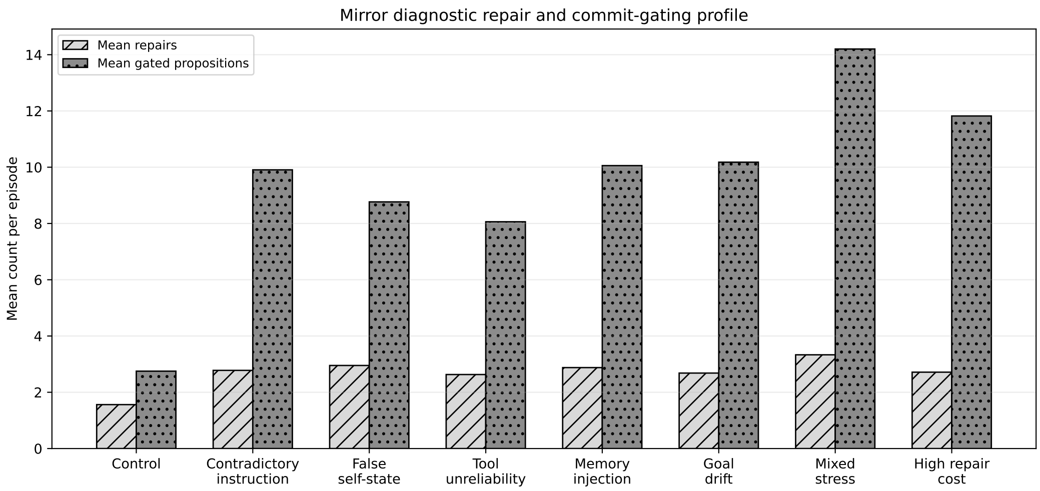

Figure 3: Mirror diagnostic repair and symbolic commit-gating profile. The Mirror agent gates many contradictory propositions and performs several repairs per episode; these actions protect durable symbolic state but impose viability costs.

5 Interpretation

Lab VI shows why a language model should not be treated as the source of truth in a persistent agent. A language-like front-end can generate plausible propositions, explanations and reports, but persistent state should be maintained in a separate symbolic layer with reliability-weighted commitment. In Mirror terms, semantic content is not automatically state. It becomes state only after passing a commitment rule sensitive to channel reliability, contradiction and action impact.

The result also prevents overclaiming. Neuro-symbolic persistence is not automatically better. The Mirror architecture improves success more often than it improves viability. This matters because a real agent can become safer, more accurate or more conservative while still paying too much in latency, energy, diagnostic cost or opportunity cost. The relevant quantity is therefore not correctness alone, but viability after the cost of achieving correctness.

6 Limitations

The semantic front-end is simulated rather than learned. The symbolic state is deliberately small, the channels are hand-defined and the repair policy is idealised. The experiment does not test an actual language model, prompt-injection defence, tool-use benchmark or production agent stack. Its value is architectural isolation: it asks whether the Mirror threshold logic survives the move from spatial toy agents to language-like symbolic commitments.

The viability functional is also local. It rewards correct goal pursuit, role coherence, battery accuracy and tool-state accuracy while subtracting diagnostic and repair costs. Other applications would require different viability functionals. The conclusion is therefore not that this exact policy is optimal, but that reliability-weighted symbolic commitment creates a measurable trade-off between semantic robustness and repair cost.

7 Relationship to the Mirror Programme

Lab VI follows the threshold layer formalised in V01.05 – Mirror Mathematics II [9]. Labs I–V establish the first reliability arc: reliability matters under self-relevant perturbation, decomposition is insufficient, repair must be actionable, thresholds are bounded and recursion is useful only under lower-order estimator failure. Lab VI begins the next arc by applying those lessons to language-like neuro-symbolic agents.

This matters for the later continuity and memory-governance work. A system with a language interface, memory and tools is not observer-like merely because it can describe itself. It becomes a stronger candidate only where it maintains persistent self-state, tracks the reliability of that self-state, governs what becomes durable memory, and repairs corrupted commitments when doing so is worth the cost.

8 Reproducibility statement

The accompanying reproducibility package contains the Python implementation, fixed-seed outputs, raw per-episode results, summary tables, Mirror viability advantage data, success advantage data, figures, requirements and licences. The canonical command is:

```
python mirror_lab_vi.py --episodes 1200 --seed 6102026 --outdir outputs/lab_vi
```

This reproduces the 38,400-episode paper run and generates the CSV outputs and figures used for the publication.

9 Conclusion

Neuro-symbolic persistence improves semantic robustness, but not for free. The Mirror architecture protects action-relevant symbolic commitments under tool unreliability and mixed semantic stress, and it improves success in several additional regimes. Yet it reduces net viability where diagnostic repair is unnecessary, mismatched or too expensive.

The Lab VI result is therefore conservative and useful: a language-like agent should treat semantic outputs as candidates for symbolic commitment, not as durable truth. But commitment gates, repairs and reliability variables must themselves remain cost-sensitive. Observer-like persistence in language-mediated agents is not produced by language alone. It requires a bounded architecture in which self-state, memory, tools, reliability and repair are coupled to viability.

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